

Used Car Price Prediction Using Regression Models

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1. Abstract

The project involved training, testing and evaluating a **Predictive regression model** to estimate used car market prices and evaluate **Image clustering performance** using Python and libraries like Pandas, NumPy, and Scikit-learn. Starting with a dataset of 4,009 records, the process included loading the provided dataset, standardizing missing or invalid string values to 'Unknown', cleaning numeric columns such as mileage and price by removing symbols, and utilizing feature engineering to calculate vehicle age from the model year. Data preprocessing further involved removing price outliers above the 99th percentile, grouping rare categorical classes into an 'others' category, and applying Label Encoding to prepare the features for training. A Random Forest Regressor was successfully developed using an 80:20 train-test split, achieving a predictive R^2 score. Additionally, the study explored unsupervised learning by applying the K-Means algorithm to the MNIST digit dataset, which achieved a macro-averaged F1 score after mapping clusters to their most frequent true labels.

2. Introduction

The project titled *“Used Car Price Prediction Using Regression Models”* has been developed to address the need for accurate and data-driven valuation of used vehicles in the automotive resale market.

Determining the appropriate price of a used car is a complex task, as it depends on multiple factors such as vehicle age, brand, mileage, fuel type, and overall condition and traditional pricing methods often lack consistency, leading to inefficiencies and reduced transparency in the secondary market.

This project aims to overcome the above-mentioned inconsistency and lack of transparency in used car pricing by training, testing and evaluating a machine learning-based predictive system that provides more accurate and data-driven price estimations. The implementation was carried out using Python as the primary programming language, along with libraries such as Pandas, NumPy, and Scikit-learn for data preprocessing and model development.

In the initial stages dataset was subjected to a series of preprocessing steps, including handling missing values, feature engineering, removal of outliers, and encoding of categorical variables to ensure suitability for machine learning algorithms. Following this, a Random Forest Regressor was implemented using an 80:20 train-test split to develop a high-accuracy price prediction model.

In parallel, the project also explored unsupervised learning by applying K-Means clustering on the MNIST digit dataset to identify inherent patterns within the data.

K-Means clustering helped in understanding how meaningful patterns and groups can be identified in unlabeled data. This concept has widespread industrial applications, including customer segmentation, recommendation systems, fraud detection, and image classification, where labeled data is often unavailable or costly to obtain.

The final phase involved a comprehensive evaluation of both approaches, where the regression model achieved an R^2 score of 0.7468, while the clustering model attained a macro-averaged F1 score of 0.8628 after appropriate label mapping.

During the initial two weeks of the internship, training was provided on several foundational topics essential for data science and machine learning. This included **Python basics** such as variables, data types, lists, loops, functions, classes, object-oriented programming (OOPs), and core data structures. Additionally, practical exposure was given to NumPy and Pandas for data manipulation and analysis. In addition, an overview of **Machine learning concepts** was covered, including two detailed sessions on data preparation, train-test splitting, model training, visualization, and performance evaluation techniques for regression, classification, and clustering tasks. The training further included introductory lectures on **Large Language Model (LLM) fundamentals**, and **communication skills** to enhance both technical and professional interaction and presentation abilities.

3. Project Objective

The primary objective of this project was to apply machine learning techniques to solve the used car price prediction problem statement provided during the internship and to gain practical exposure to both supervised and unsupervised learning methodologies.

The specific objectives of the project are as follows:

1. To analyse the structured vehicle dataset provided by the institute and identify the factors influencing used car prices.
2. To perform data preprocessing and feature engineering, including handling missing values, removing outliers, cleaning numerical attributes, and encoding categorical variables to prepare the data for machine learning models.
3. To develop and evaluate a Random Forest Regressor using an 80:20 train-test split for accurate prediction of used car prices and assess its performance using appropriate evaluation metrics.
4. To gain hands-on experience in the complete machine learning pipeline, including data visualization, model training, testing, validation, and performance evaluation.
5. To explore unsupervised learning concepts by implementing K-Means clustering on the MNIST digit dataset and demonstrating the ability of clustering algorithms to discover inherent patterns in unlabelled data.

Note: The dataset and project tasks were provided by the institute as part of the internship training program. No independent survey was conducted, and no hypothesis testing was performed in this project.

4. Methodology

The project was carried out using a structured vehicle dataset provided by the institute as part of the internship program. The primary objective was to develop a supervised machine learning model for predicting used car prices. Additionally, practical exposure to unsupervised learning techniques was gained through the implementation of K-Means clustering on the MNIST handwritten digit dataset. Python was used as the primary programming language, while libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn were utilized for data processing, visualization, model development, and evaluation. The implementation was performed using Google Collab and Visual Studio Code (VS Code), which provided an efficient environment for coding, experimentation, and

analysis. The project was carried out using a structured vehicle dataset and an accompanying IPython Notebook (. ipynb) containing the project questions and tasks, both provided by the institute as part of the internship program.

Training and Project Assignment

Before the commencement of the project, a two-week training program was conducted as part of the internship. The training covered Python fundamentals, including data types, variables, lists, loops, functions, data structures, classes, and object-oriented programming (OOP). Practical sessions on NumPy and Pandas were also conducted for data manipulation and analysis.

In addition, the training introduced machine learning concepts such as data preprocessing, train-test splitting, model training, visualization, and performance evaluation for regression, classification, and clustering tasks. Lectures on machine learning fundamentals, Large Language Models (LLMs), and communication skills were also delivered. Upon completion of the training, the project dataset, notebook containing the project questions, and implementation guidelines were provided for practical application of the concepts learned.

The methodology for project completion followed a systematic workflow comprising data collection, data understanding, preprocessing, feature engineering, exploratory data analysis, model development, evaluation, and interpretation of results.

1. Data Collection

The dataset (used_cars.csv) used for the project was provided by the institute and consisted of 4,009 records containing various attributes related to used vehicles. Along with the dataset, an IPython Notebook (. ipynb) containing the project questions and tasks was also provided as part of the internship assignment. Since the dataset and project requirements were supplied by the institute, no independent survey was conducted and no primary data collection was required.

2. Data Understanding and Exploration

The dataset was explored through the activities covered in Questions 2-3 of the provided notebook, which involved inspecting the dataset structure, identifying missing values, examining feature characteristics, and understanding distribution of variables. This helped in gaining a better understanding of the data and identifying issues that needed to be addressed during preprocessing.

3. Data Cleaning and Preprocessing

Several preprocessing steps were carried out to improve data quality and prepare the dataset for machine learning algorithms:

- Standardization of missing and invalid values.
- Cleaning of numerical columns by removing currency symbols, commas, and other non-numeric characters.
- Conversion of data into appropriate numerical formats.
- Creation of a new feature, car_age, from the model year of the vehicle.
- Handling of categorical variables through Label Encoding.
- Grouping of low-frequency categories where required.
- Detection and removal of outliers using percentile-based techniques to reduce the

impact of extreme values.

4. Data Visualization and Analysis

Data visualization was used to present and interpret the results obtained from the machine learning models. For the used car price prediction task (Question 10), visualizations such as Actual vs Predicted Price plots were used to evaluate the performance of the Random Forest Regressor. Similarly, for the K-Means clustering task on the MNIST dataset (Question 11, 12, 13) visual representations of clusters could be done to observe how handwritten digit images were grouped by the algorithm.

These visualizations were not done in notebook as they were not explicitly mentioned to be performed but one of the visualizations for car price prediction is done separately and added in result section as it helped in understanding model behavior and assessing the effectiveness of the developed solutions.

5. Model Development

The project involved two machine learning approaches:

a) Supervised Learning – Used Car Price Prediction

A Random Forest Regressor was implemented to predict used car prices. The dataset was divided into training and testing sets using an 80:20 split, where 80% of the data was used for model training and 20% was reserved for testing and validation.

The Random Forest Regressor was selected because of its ability to handle complex relationships among variables, reduce overfitting through ensemble learning, and provide robust predictive performance on structured datasets.

b) Unsupervised Learning – K-Means Clustering

To gain practical exposure to unsupervised learning, K-Means clustering was applied to the MNIST handwritten digit dataset. The objective was to identify inherent patterns and group similar images without using label information during training. This exercise demonstrated how clustering algorithms can discover meaningful structures in unlabelled data.

6. Model Evaluation and Validation

1. The performance of the Random Forest Regressor was evaluated using the following metrics:
 - R^2 Score
 - Mean Squared Error (MSE)
 - Mean Absolute Error (MAE)

The final regression model achieved an R^2 score of 0.7468, indicating good predictive capability.

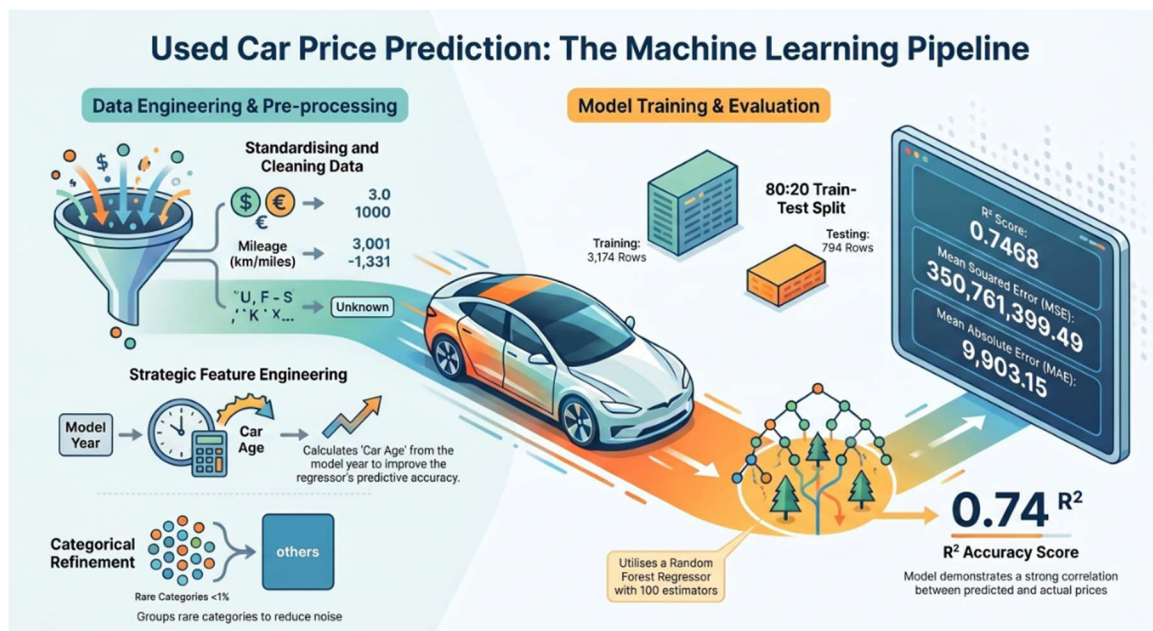
2. For the K-Means clustering model, cluster labels were mapped to the corresponding true digit labels, and performance was evaluated using the Macro-Averaged F1 Score.

The clustering model achieved a Macro F1 Score of 0.8628 after label mapping.

7. Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook
- Google collab notebook
- Excel spreadsheet
- VS code
- Notebook LM
- GitHub

Figure: Machine Learning workflow for Used Car Price Prediction



Source: Workflow diagram generated using Google NotebookLM based on the project methodology and implementation steps.

GitHub Repository

The complete source code developed during the project was uploaded to GitHub and the repository link is included here for reference.

GitHub Link: https://github.com/shriya829/15-Used_Car_Price_Prediction_Summer_2026

5. Data Analysis and Results

This section presents the key findings obtained from the data analysis and machine learning tasks performed during the internship. The results are organized into descriptive analysis and machine learning model evaluation. Since the project was based on institute-provided datasets and focused on predictive modeling and clustering, no survey was conducted and no hypothesis testing was done.

1. Descriptive Analysis

The used car dataset consisted of 4,009 records containing information related to vehicle characteristics and market prices. Initial analysis revealed the presence of missing values, inconsistent formatting in numerical fields, and outliers that required preprocessing before model development.

The following observations were made during data analysis -:

Analysis Aspect	Observation
Dataset Size	4,009 vehicle records
Missing Values	Present and standardize during preprocessing
Numerical Data	Cleaned by removing currency symbols and commas
Feature Engineering	New feature car_age created from manufacturing year
Categorical Variables	Processed using Label Encoding
Outliers	Identified and removed using percentile-based methods

Relevant code snippets from the corresponding project questions have been included to illustrate how the results were generated.

```
# Loading the dataset into a pandas dataframe
df = pd.read_csv("used_cars.csv")

# Shuffling the dataframe
df1 = df.sample(frac=1, random_state=42)

# To Print a Series showing the count of nulls per column
print("The number of missing values present in each column are : ")
print(df1.isnull().sum())
```

```

# To perform replacements to standardize missing data
df1[obj_cols.columns] = obj_cols.apply(
    lambda x : x.replace('-', 'Unknown') # x is a series (a single column)
                .replace('nan', 'Unknown') # replace specific text values '-' and 'nan' with 'Unknown'
                .replace(r'^\s*$', 'Unknown', regex=True), # replace strings that are either empty or contain only whitespaces,
    axis=0 # apply the lambda to each column(Series) in the obj_cols
)

# With the above code , the code cell executed without errors , standardizing the text data.

# creating new column by calculating car age from model_year using 2026 as current year
df1['car_age'] = 2026 - df1['model_year']

# Dropping the original model_year column
df1 = df1.drop(columns=['model_year'])

# Find the 99th percentile price
price_99 = df1['price'].quantile(0.99)

# Remove price outliers by keeping the rows with price below the 99th percentile
df1 = df1[df1['price'] < price_99].copy()

# Applying label encoding to convert all remaining categorical(object) columns into numerical values
le = LabelEncoder()
categorical_cols = df1.select_dtypes(include=['object', 'string']).columns

for col in categorical_cols:
    df1[col] = le.fit_transform(df1[col].astype(str))

```

2. Supervised Learning Results – Used Car Price Prediction

A Random Forest Regressor was implemented using an 80:20 train-test split to predict used car prices. The model demonstrated good predictive performance after preprocessing and feature engineering.

The following performance summary was obtained: -

Evaluation Metric	Value
R ² Score	0.7468
Mean Squared Error (MSE)	350,761,399.49
Mean Absolute Error (MAE)	9,903.15

The obtained R² score indicates that the model was able to explain approximately 74.68% of the variation in used car prices, demonstrating satisfactory predictive capability.

Relevant code snippets from the corresponding project questions have been included to illustrate how the results were generated.

```

# Splitting the features and target into training and testing dataset using 80:20 ratio
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

# Training the model on training data.
rf_model = RandomForestRegressor(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)

# Predicting the price for the test data.
y_prediction = rf_model.predict(X_test)

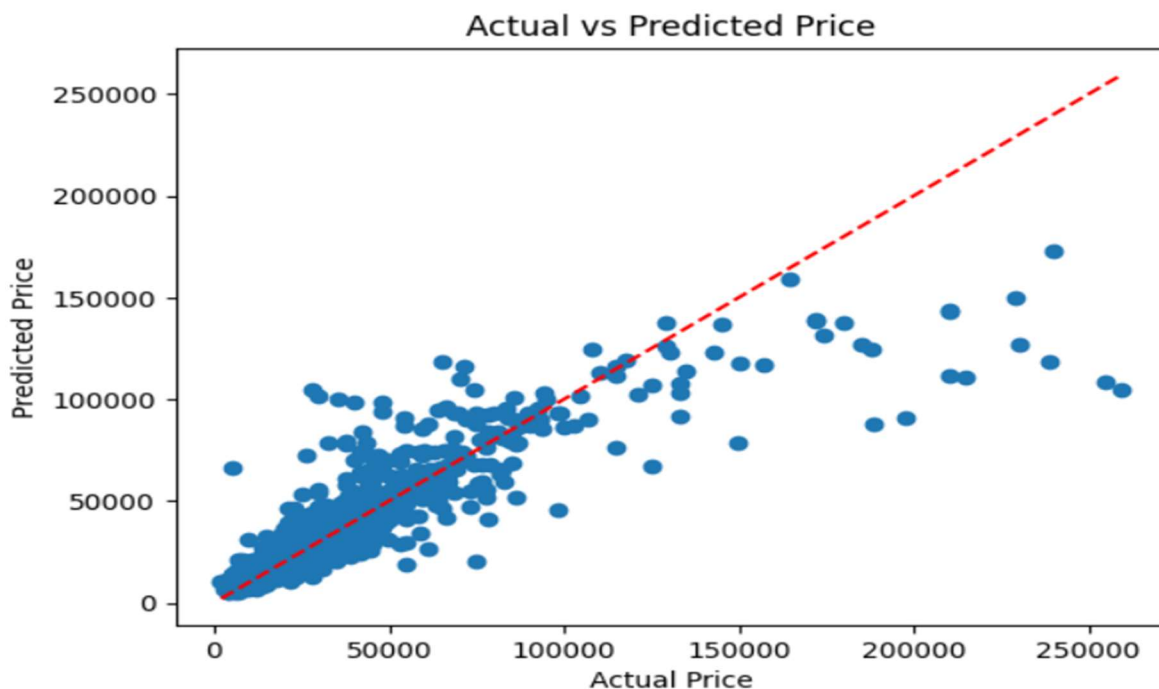
# Evaluating the model
mae = mean_absolute_error(y_test, y_prediction)
mse = mean_squared_error(y_test, y_prediction)
r2 = r2_score(y_test, y_prediction)

# Printing the results obtained from the model
print("R2 Score:", r2)
print("Mean squared error (MSE):", mse)

```

The Data Visualisation done for better understanding is added below along with the code snippet used to generate it :-

The Actual vs Predicted Price plot demonstrates that the Random Forest Regressor was able to learn the relationship between vehicle attributes and market price effectively. Most data points lie close to the ideal prediction line, indicating good predictive performance. Although larger deviations are observed for some high-priced vehicles, the overall trend confirms that the model provides reliable price predictions, which is further supported by the achieved R² score.



```
import matplotlib.pyplot as plt

plt.scatter(y_test, y_prediction)
min_val = min(y_test.min(), y_prediction.min())
max_val = max(y_test.max(), y_prediction.max())
plt.plot([min_val, max_val], [min_val, max_val], 'r--')
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted Price")
plt.show()
```

3. Unsupervised Learning Results – K-Means Clustering

K-Means clustering was applied to the MNIST handwritten digit dataset to identify natural groupings in image data without using labels during training.

Evaluation Metric	Value
Number of Clusters	10
Dataset	MNIST Handwritten Digits
Macro-Averaged F1 Score	0.8628

Cluster labels were mapped to the most frequent true digit labels within each cluster for evaluation purposes. The obtained Macro F1 Score of 0.8628 indicates effective grouping of similar handwritten digits.

Relevant code snippets from the corresponding project questions have been included to illustrate how the results were generated for F1 score.

```
# Importing required libraries
from sklearn.metrics import f1_score
import numpy as np

# Mapping the labels
mapped_labels = np.zeros_like(kmeans_labels) # Creating an array to store the mapped true labels

for cluster in np.unique(kmeans_labels): #loop through each unique cluster
    cluster_mask = (kmeans_labels == cluster) #Selecting all the samples belonging to current cluster
    most_common_label = np.bincount(y[cluster_mask]).argmax() # Finding the most frequent true label in this cluster
    mapped_labels[cluster_mask] = most_common_label # Mapping the most frequent true label to each sample in cluster

# Calculating macro-averaged F1 score
macro_f1 = f1_score(y, mapped_labels, average='macro')

# Printing the macro-averaged F1 score
print("The calculated F1 score of the clustering:", macro_f1)
```

6. Conclusion

Based on the analysis and results obtained, following conclusions can be drawn: -

- The project demonstrated the effective application of both supervised and unsupervised machine learning techniques.
- The Random Forest Regressor proved to be a reliable model for used car price prediction, achieving an **R² score of 0.7468** and a **Mean Absolute Error (MAE) of 9,903.15**, indicating good predictive performance.
- The results also highlighted the importance of data preprocessing, feature engineering, and outlier removal in improving model accuracy.
- Additionally, K-Means clustering on the MNIST dataset achieved a **Macro-Averaged F1 Score of 0.8628** after label mapping, demonstrating its ability to identify meaningful patterns in unlabeled data.

Overall, the project provided practical experience in data preprocessing, model development, and performance evaluation using both supervised and unsupervised learning approaches.

7. APPENDICES

Appendix A: References

The following resources were referred to during the internship and project implementation:

1. Scikit-learn Documentation – Machine Learning in Python.
2. Pandas Documentation – Data Analysis and Manipulation Tools.
3. NumPy Documentation – Numerical Computing in Python.
4. Matplotlib Documentation – Data Visualization Library.
5. Python Official Documentation.
6. Lecture materials and notebooks provided during the internship training program.
7. MNIST Handwritten Digit Dataset Documentation.
8. Used Car Dataset provided by the institute as part of the internship project.
9. Project Notebook (.ipynb) containing the project questions and implementation guidelines provided by the institute.

Appendix B: GitHub Repository

The complete source code developed during the internship project has been uploaded to GitHub.

GitHub Repository Link: [https://github.com/shriya829/15-](https://github.com/shriya829/15-Used_Car_Price_Prediction_Summer_2026)

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